Module 0 — Read 0

Welcome to STAT 109

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Welcome to STAT 109: Introductory Biostatistics at Cal Poly Humboldt.

At its core, **statistics** is the *science of using data to answer questions and make decisions*. Throughout this course, we will learn how to use data to better understand the world around us and to make decisions in the presence of uncertainty.

This course is organized around a simple idea: statistics is best learned by doing (with a bit of thinking). Rather than spending most of the time watching lectures or memorizing formulas, you will formulate questions, collect and analyze data, build statistical models, evaluate evidence, and communicate conclusions. By the end, you will have completed five projects that demonstrate your ability to use statistics to answer questions that are meaningful to you or someone else in class.

The Statistical Process

Although the tools change from project to project, most statistical investigations follow the same basic process.

Step 1: Ask a Well-Defined Question

Every investigation begins with a question.

For example, one I've often heard is

- Does listening to music while studying result in higher grades?

An well-defined question is specific, measurable, and answerable with data. To make this a well-defined question, think about exactly what we'd need to specify before data could be used: *who is this question about? what sort of music? How long? Higher grade in what?*

Step 2: Make a Plan to Collect, Analyze and Conclude

Statistical investigations that are efficient and answer the research question are designed before data are collected. Before collecting data, we need a plan.

We decide:

- what data to collect,
- how we will analyze the data,
- how we will make a conclusion based on the analysis.

Step 3: Collect Data

We gather information from the world we observe.

This might involve:

- surveys,
- experiments,
- observations,
- measurements,
- existing datasets.

Step 4: Analyze Data

We summarize, visualize, and model the data.

This is where statistical tools help us identify patterns, estimate quantities, test claims, or make predictions. It often involves a computer.

Step 5: Draw Conclusions

Finally, we connect the results back to the original question.

A good conclusion:

- answers the question,
- acknowledges uncertainty,

- explains limitations,
- communicates findings clearly.

Throughout this course, we will practice all five steps.

Three Worlds

Statistics takes place in three worlds.

The World We Observe

This is the world of things we can directly see, measure, and record. For example,

- STAT 109 student responses in a discussion thread in Summer of 2026,
- hours of sleep I get this week,
- wildlife sightings in my backyard today,
- high temperature in Eureka on June 1, 2026,
- size of muffins at Costco on June 1, 2026 (aren't they smaller than they used to be?).

The World We Imagine

This is the world of statistical ideas and models where the amount of data is unlimited by any constraints on what I can actually collect.

Examples:

- responses in a discussion thread from all students in all of time
- hours of sleep I get tomorrow night,
- wildlife sightings in my city,
- rise in temperature caused by human driving red cars instead of white cars,
- mean difference size of muffins at Costco in 2026 and 2006.

Don't tell the other students this, but mathematics turns out to be quite useful in imagining worlds that are useful for connecting to the world we can observe. We will do some mathematics, especially in Module 1, but only the useful sort—not the just-for-fun look at how $e^{i\pi} = -1$ sort. Disappointed? I know, but cheer up: there is one place in the course where π turns out to be useful, if not exactly fun.

The World of Computers

Computers help us move between what we observe and what we imagine.

They allow us to:

- organize data,
- create visualizations,
- fit models,
- perform calculations,
- communicate results.

Learning statistics is largely the process of learning how to move among these three worlds. Sounds hard, right? Especially when you consider that each world has it's own language! Well, I'm not going to lie, it is hard. But that's why you're taking a class with a helpful professor like me!

The Five Projects

The course is organized around five portfolio projects.

Project 1: Testing a Claim

Is what we observed surprising?

Project 1 — Testing a Claim

Project 2: Estimating the Truth

What might the true population value be?

Project 2 — Estimating the Truth

Project 3: Discovering Patterns

What patterns are present in a dataset?

Project 3 — Discovering Patterns

Project 4: Designing Fair Comparisons

How can we make a fair comparison?

Project 4 — Designing Fair Comparisons

Project 5: Looking Ahead

What might happen?

Project 5 — Looking Ahead

By the end of the course, you will have a portfolio of projects that you can to share with your family, friends and potential employers.

Google Colab and R

All of our work will take place in Google Colab.

Google Colab is a cloud-based notebook environment that runs entirely in your web browser. You do not need to install any special software.

Inside Google Colab, we will use the R programming language.

A notebook contains three kinds of information.

Text

Text cells are used to communicate with people.

You can use them to:

- explain your thinking,
- answer questions,
- record notes for your future self,
- communicate conclusions to others.

Think of text cells as pages in a lab notebook.

Code

Code cells contain instructions for the computer.

For example, the instruction asks the computer (in the R language) to compute the mean of the numbers 7, 2 and 3:

```
mean(c(7, 2, 3))
```

If I were doing it by hand, I've have written $\frac{7+2+3}{3}$. I could tell this to a computer, but the language of R is more efficient when there are lots of numbers OR I don't want have to remember the math formula for what I want calculated (remember we're skipping the for-fun math).

Throughout the course, you will learn to read, modify, and create pieces of code in the language R (yes, this is one of those new languages you'll learning!).

Output

Output is what the computer produces after running code.

Output might include:

- numbers,
- tables,
- graphs,
- error messages,
- statistical results.

One of the most important skills in this course is learning how to interpret output and connect it back to the original question.

What You'll Do This Module

This module is designed to help you get comfortable with the course structure before we begin our first project.

You will:

1. Complete a short lab introducing Google Colab and R — demo lab · exercise lab.
2. Submit the lab for participation credit. Think of this as free points and a chance to practice. I will provide feedback to help you get started.
3. Introduce yourself to your classmates in the discussion board (Canvas Module 0).
4. Complete a short practice quiz covering:
   - the definition of statistics,
   - the five steps of the statistical process,
   - the three worlds,
   - basic notebook concepts.

   Quiz 0 (Start Here) · practice on the site; graded version on Canvas.
5. Getting on the first step of the statistical process by submit three research questions — Quiz 0B (Question Ideas) on Canvas.

Also read the syllabus - it's got all the fine print.

Looking Ahead: Research Questions

Our first project begins with Step 1 of the statistical process: asking a good question.

For your final activity in this module, you will submit three possible research questions involving a comparison between two groups.

Examples:

- Do people who exercise regularly report higher energy levels than people who do not?
- Do students who live on campus get more sleep than students who live off campus?
- Do grandmothers who own a pet report lower stress levels than grandmothers who do not?
- Has the volume of muffins at Costco decreased over the last two decades?
- What is the chance of finding a parking space on campus at 8:15 vs. 8:55.

The goal is not to have the perfect question yet. The goal is to think of some questions that you actually care about answering.

Welcome to STAT 109 - I'm glad you're here and am excited to support you in this three world adventure.